Chaos-Based Regime Detection in Financial Markets: A Hidden Markov Model Approach with Nonlinear Dynamical Metrics

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Abstract

We propose a regime detection framework for financial markets that integrates nonlinear dynamical metrics—the maximal Lyapunov exponent (Rosenstein's method), the Hurst exponent (detrended fluctuation analysis), and permutation entropy (Bandt–Pompe)—into a three-state Gaussian Hidden Markov Model. Applied to daily returns of 20 major U.S. equities from 2010 to 2025, the model identifies three dynamically distinct regimes characterized as *chaotic* (high Lyapunov exponent, elevated complexity), *trending* (high Hurst exponent, persistent dynamics), and *mean-reverting* (low Hurst exponent, anti-persistent behavior). We validate regime assignments through iterated amplitude-adjusted Fourier transform (IAAFT) surrogate testing, finding that observed Lyapunov exponents are broadly consistent with stochastic linear processes, while permutation entropy reveals significant nonlinear structure in select assets. A regime-conditioned trading strategy achieves positive risk-adjusted returns in 65% of assets tested, with meaningful drawdown reduction for high-volatility equities. Our findings suggest that while deterministic chaos per se is not robustly detectable in daily equity returns, the dynamical complexity decomposition provided by chaos-theoretic metrics offers a richer characterization of market regimes than volatility-based approaches alone. This framework provides practitioners with an enhanced toolkit for regime-dependent portfolio construction and risk management.

Keywords: Regime detection, Hidden Markov Models, Lyapunov exponent, Hurst exponent, permutation entropy, chaos theory, financial markets, nonlinear dynamics

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1 Introduction

Financial markets exhibit complex, time-varying dynamics that challenge both academic theory and practical risk management. The observation that market behavior alternates between distinct "regimes"—periods of calm trending, volatile mean-reversion, and apparently unpredictable turbulence—has motivated extensive research in regime-switching models [Hamilton, 1989, Guidolin, 2011, Kritzman et al., 2012]. Yet the dominant approaches to regime identification rely predominantly on first and second moment characteristics (returns and volatility), potentially overlooking deeper structural features of market dynamics.

The application of nonlinear dynamics and chaos theory to financial markets has a rich but contested history. Early contributions by Scheinkman and LeBaron [1989] and Hsieh [1991] explored whether stock returns exhibit deterministic chaos, while Peters [1994] and Mandelbrot [1997] advanced fractal perspectives on market behavior. A central challenge in this literature is distinguishing genuine deterministic chaos from high-dimensional stochastic processes—a distinction that requires careful statistical methodology [Theiler et al., 1992, Kantz and Schreiber, 2004].

This paper contributes to the intersection of nonlinear dynamics and financial econometrics by proposing a *chaos-augmented Hidden Markov Model* (HMM) framework for regime detection. Our approach enriches the standard HMM observation vector with three chaos-theoretic metrics computed on rolling windows of daily returns:

1. **Maximal Lyapunov exponent** $\lambda_{\max}$, estimated via Rosenstein et al. [1993]'s method, which quantifies sensitive dependence on initial conditions—the hallmark of deterministic chaos.

2. **Hurst exponent** H, estimated via detrended fluctuation analysis [DFA; Peng et al., 1994], which characterizes long-range dependence and persistence structure.

3. **Permutation entropy** H_{PE}, following Bandt and Pompe [2002], which measures the complexity of ordinal patterns in the time series.

The three-state HMM maps latent states to economically interpretable regimes: *chaotic* (high dynamical complexity, elevated Lyapunov exponent), *trending* (persistent dynamics, high Hurst exponent), and *mean-reverting* (anti-persistent dynamics, low Hurst exponent). We validate our regime assignments using iterated amplitude-adjusted Fourier transform (IAAFT) surrogate testing [Schreiber and Schmitz, 1996] to assess statistical significance of detected nonlinear structure.

We apply this framework to 20 major U.S. equities spanning diverse sectors over the period January 2010 to March 2025. Our empirical analysis yields several key findings:

- The HMM identifies three dynamically distinct regimes across all assets, with meaningful separation in chaos metric profiles and economically interpretable characteristics.
- Surrogate testing indicates that maximal Lyapunov exponents in daily equity returns are broadly consistent with linear stochastic processes, while permutation entropy shows significant departure from linearity in select assets.
- The identified regimes exhibit distinct return and volatility characteristics, with statistically significant differences in next-period returns for high-volatility assets.
- A regime-conditioned trading strategy achieves positive Sharpe ratios for 13 of 20 assets, with notable drawdown reduction in volatile stocks relative to buy-and-hold.

These findings contribute to the debate on chaos in financial markets by providing rigorous, multi-asset evidence with proper surrogate-based validation. Our methodological contribution lies in demonstrating that chaos-theoretic metrics, even when deterministic chaos is not conclusively established, provide valuable regime characterization beyond what volatility alone captures.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the mathematical framework for each metric and the HMM model specification. Section 4 describes the data. Section 5 reports empirical results. Section 6 provides robustness checks. Section 7 discusses implications, and Section 8 concludes.

2 Literature Review

2.1 Chaos and Nonlinear Dynamics in Finance

The investigation of chaotic dynamics in financial markets began in earnest in the late 1980s. Scheinkman and LeBaron [1989] applied the correlation dimension test [Grassberger and Procaccia, 1983] to stock returns, finding evidence of low-dimensional deterministic structure. Hsieh [1991] conducted a comprehensive analysis of daily stock returns using the BDS test and Lyapunov exponents, concluding that while returns exhibit significant nonlinearity, the evidence for low-dimensional chaos was inconclusive. Brock et al. [1991] provided a systematic treatment of statistical tools for detecting nonlinear dynamics in economic data.

The fractal market hypothesis [Peters, 1994, Mandelbrot, 1997] offered an alternative perspective, emphasizing the role of long-range dependence and self-similar scaling in financial time series. Lo [1991] introduced a modified R/S statistic to test for long memory in stock returns, finding mixed evidence. More recently, Di Matteo et al. [2005]

applied DFA to financial data, documenting time-varying Hurst exponents across different market conditions.

The application of permutation entropy to financial data is more recent. Zunino et al. [2009] used permutation-based measures to analyze stock market inefficiency, while Zunino et al. [2012] explored the complexity-entropy causality plane as a tool for characterizing financial dynamics. These studies demonstrate that ordinal pattern analysis captures aspects of time series structure invisible to traditional linear methods.

A persistent challenge in this literature is the surrogate data methodology for distinguishing deterministic chaos from stochastic nonlinearity [Theiler et al., 1992]. Schreiber and Schmitz [1996] introduced the IAAFT algorithm, which generates surrogates preserving both the power spectrum and amplitude distribution of the original series, providing a stringent null hypothesis for nonlinearity testing.

2.2 Regime-Switching Models in Finance

Hidden Markov Models and Markov-switching models have been widely applied in financial econometrics since the seminal work of Hamilton [1989]. Ang and Bekaert [2002] applied regime-switching models to interest rates, while Hardy [2001] developed regime-switching models for long-term stock returns. Guidolin [2011] provides a comprehensive survey of Markov-switching applications in empirical finance.

Kritzman et al. [2012] demonstrated the practical value of regime detection for dynamic asset allocation, showing that regime-aware strategies can significantly reduce portfolio drawdowns. The standard approach uses returns and volatility as observables, with latent states typically capturing "bull" and "bear" markets or "high" and "low" volatility regimes.

2.3 Our Contribution

To our knowledge, no prior study has systematically combined Lyapunov exponent, Hurst exponent, and permutation entropy within an HMM framework for financial regime detection, with formal surrogate-based validation across a large cross-section of assets. Our work bridges the chaos-in-finance literature with modern regime-switching econometrics, providing a rigorous empirical assessment of whether chaos-theoretic metrics enhance regime identification beyond standard volatility-based approaches.

3 Methodology

3.1 Data Preprocessing

Let P_t denote the adjusted closing price of an asset at time t. We compute log-returns as:

$$r_t = \ln P_t - \ln P_{t-1} \tag{1}$$

Returns are winsorized at $\pm 10\sigma$ to mitigate the influence of extreme outliers on phase-space reconstruction. Rolling features are computed on overlapping windows of varying lengths, aligned at the window endpoint to avoid look-ahead bias.

3.2 Maximal Lyapunov Exponent (Rosenstein's Method)

The maximal Lyapunov exponent $\lambda_{\max}$ quantifies the average exponential rate of divergence of initially nearby trajectories in phase space. A positive $\lambda_{\max}$ is a necessary condition for deterministic chaos.

Definition 1 (Lyapunov Exponent)**.** *For a dynamical system with trajectory* $\mathbf{x}(t)$*, the maximal Lyapunov exponent is defined as:*

$$\lambda_{\max} = \lim_{t\to\infty} \lim_{\|\delta\mathbf{x}(0)\|\to 0} \frac{1}{t} \ln \frac{\|\delta\mathbf{x}(t)\|}{\|\delta\mathbf{x}(0)\|} \tag{2}$$

where $\delta\mathbf{x}(t)$ *is an infinitesimal perturbation to the trajectory.*

For a scalar time series, we first reconstruct the phase space using Takens' delay embedding theorem [Takens, 1981]:

$$\mathbf{x}_i = [x(t_i),\ x(t_i+\tau),\ \ldots,\ x(t_i+(m-1)\tau)] \in \mathbb{R}^m \tag{3}$$

where m is the embedding dimension and τ is the time delay.

Rosenstein's algorithm [Rosenstein et al., 1993] proceeds as follows:

Algorithm 1 Rosenstein's Maximal Lyapunov Exponent

Require: Time series $\{x_t\}_{t=1}^{N}$, embedding parameters (m, τ), Theiler window w_T
Construct embedded vectors $\{\mathbf{x}_i\}_{i=1}^{M}$ where $M = N - (m-1)\tau$
for each $\mathbf{x}_i$ **do**
  Find nearest neighbor $\mathbf{x}_{j^*(i)}$ with $|i - j^*(i)| > w_T$
end for
for each time step $k = 0, 1, \ldots, k_{\max}$ **do**
  Compute $d_i(k) = \|\mathbf{x}_{i+k} - \mathbf{x}_{j^*(i)+k}\|$
  Compute $\langle \ln d(k) \rangle = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \ln d_i(k)$
end for
Estimate $\lambda_{\max}$ as the slope of $\langle \ln d(k) \rangle$ vs. $k\Delta t$ in the initial linear region
return $\lambda_{\max}$

The Theiler window $w_T \geq m\tau$ excludes temporally correlated neighbors to avoid biased estimates [Theiler et al., 1992]. We use a KD-tree for efficient nearest-neighbor queries and fit the slope over the initial $k_{\max}/4$ time steps, where the divergence curve is approximately linear.

Parameters: embedding dimension $m = 3$, time delay $\tau = 1$, Theiler window $w_T = 3$, maximum divergence steps $k_{\max} = 40$, rolling window $W_{\text{long}} = 250$ trading days.

3.3 Hurst Exponent via Detrended Fluctuation Analysis

The Hurst exponent H characterizes the scaling behavior and long-range dependence of a time series. We employ DFA [Peng et al., 1994], which is robust to non-stationarity.

Given returns $\{r_t\}$, we construct the integrated profile:

$$Y(i) = \sum_{k=1}^{i} (r_k - \bar{r}), \quad i = 1, \ldots, N \tag{4}$$

For each box size s, the profile is divided into $\lfloor N/s \rfloor$ non-overlapping segments. Within each segment v, a polynomial trend of order q is fitted and subtracted, yielding the detrended fluctuation function:

$$F(s) = \left[\frac{1}{\lfloor N/s \rfloor} \sum_{v=1}^{\lfloor N/s \rfloor} \frac{1}{s} \sum_{i=1}^{s} \left(Y_{v,i} - \hat{Y}_{v,i}^{(q)} \right)^2 \right]^{1/2} \tag{5}$$

The Hurst exponent is the slope of $\log F(s)$ versus $\log s$:

$$F(s) \propto s^H \quad \Rightarrow \quad H = \frac{d \log F(s)}{d \log s} \tag{6}$$

The interpretation is:

- $H \approx 0.5$: uncorrelated random walk (efficient market)

- $H > 0.5$: persistent (trending) behavior
- $H < 0.5$: anti-persistent (mean-reverting) behavior

Parameters: polynomial order $q = 1$ (DFA-1), box sizes logarithmically spaced from $s_{\min} = 4$ to $s_{\max} = W_{\text{med}}/4$, rolling window $W_{\text{med}} = 125$ trading days.

3.4 Permutation Entropy

Permutation entropy [Bandt and Pompe, 2002] measures the complexity of a time series based on ordinal patterns. For embedding dimension m and delay τ, each subsequence $\left[x_t, x_{t+\tau}, \ldots, x_{t+(m-1)\tau}\right]$ defines a permutation pattern $\pi \in S_m$ through its rank ordering.

The permutation entropy is:

$$H_{PE} = -\sum_{\pi \in S_m} p(\pi) \ln p(\pi) \tag{7}$$

where $p(\pi)$ is the relative frequency of pattern π. The normalized permutation entropy is:

$$h_{PE} = \frac{H_{PE}}{\ln(m!)} \tag{8}$$

satisfying $0 \leq h_{PE} \leq 1$, where $h_{PE} = 1$ corresponds to a completely random process and lower values indicate more deterministic or regular structure.

Parameters: $m = 5$ (yielding $5! = 120$ possible patterns), $\tau = 1$, rolling window $W_{\text{med}} = 125$ trading days.

3.5 Hidden Markov Model Specification

We employ a three-state Gaussian HMM [Rabiner, 1989] with observation vector:

$$\mathbf{y}_t = \left[r_t,\ \sigma_t^{(\text{short})},\ \lambda_t,\ H_t,\ h_{PE,t}\right]^\top \in \mathbb{R}^5 \tag{9}$$

where r_t is the daily log-return, $\sigma_t^{(\text{short})}$ is the 30-day rolling volatility, λ_t is the rolling Lyapunov exponent, H_t is the rolling Hurst exponent, and $h_{PE,t}$ is the rolling normalized permutation entropy.

The model is specified by:

- Initial state distribution: $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3)$
- Transition matrix: $\mathbf{A} = [a_{ij}]$ where $a_{ij} = P(s_t = j \mid s_{t-1} = i)$
- Emission distributions: $\mathbf{y}_t \mid s_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ with state-specific mean $\boldsymbol{\mu}_k$ and full covariance $\boldsymbol{\Sigma}_k$

Parameters are estimated via the Expectation-Maximization (EM) algorithm [Baum et al., 1970], and the most likely state sequence is decoded using the Viterbi algorithm [Viterbi, 1967]. Features are standardized (zero mean, unit variance) prior to model fitting.

3.6 State-to-Regime Mapping

After HMM training, we assign economically interpretable labels to latent states based on state-conditional feature means:

1. **Chaotic regime:** The state s^* with the highest mean Lyapunov exponent: $s^* = \arg\max_k \bar{\lambda}_k$
2. **Trending regime:** Among remaining states, the one with the highest mean Hurst exponent
3. **Mean-reverting regime:** The remaining state, characterized by the lowest mean Hurst exponent

This mapping is validated through surrogate testing (Section 3.7) and cross-sectional consistency.

3.7 Surrogate Testing for Statistical Validation

To assess whether observed chaos metrics reflect genuine nonlinear structure beyond what a linear stochastic process could produce, we employ IAAFT surrogates [Schreiber and Schmitz, 1996].

The IAAFT algorithm generates surrogates that preserve both the power spectrum (linear autocorrelation) and the amplitude distribution (marginal density) of the original series. The null hypothesis is:

$$H_0 : \text{The time series is a monotonic nonlinear transformation of a linear Gaussian process} \tag{10}$$

For each asset, we generate $K = 19$ surrogates and compute the test statistic (Lyapunov exponent or permutation entropy) on each. The p-value is:

$$p = \frac{\#\{T_{\text{surr}} \geq T_{\text{obs}}\} + 1}{K + 1} \tag{11}$$

for the Lyapunov exponent (one-sided test for significant positive divergence) and analogously for permutation entropy (testing for significantly lower complexity than surrogates).

3.8 Regime-Based Trading Strategy

We implement a simple regime-conditioned strategy to assess the economic significance of detected regimes:

- **Trending regime:** Long position (full allocation)
- **Mean-reverting regime:** Contrarian signal—long if $r_t < -0.005$, short if $r_t > 0.005$, flat otherwise
- **Chaotic regime:** Flat (cash position) to avoid unpredictable periods

Positions are established at time t based on the regime classification and held for one day. Transaction costs of 5 basis points per trade are applied. The benchmark is a buy-and-hold strategy.

4 Data

4.1 Sample Description

We analyze daily adjusted closing prices for 20 U.S. equities spanning multiple sectors, downloaded from Yahoo Finance. The sample period is January 4, 2010 to March 18, 2025, providing approximately 3,824 trading days per asset.

Table 1 lists the assets in our sample. The selection covers large-cap equities across technology (AAPL, MSFT, GOOGL, AMZN, NVDA, META), financials (JPM, BAC, V, MA), healthcare (JNJ, PFE, UNH), consumer (WMT, DIS, HD, PG), energy (XOM), and diversified indices (SPY), as well as high-volatility names (TSLA).

Table 1: Assets Analyzed

Ticker	Name	Sector
SPY	SPDR S&P 500 ETF	Index
AAPL	Apple Inc.	Technology
MSFT	Microsoft Corp.	Technology
GOOGL	Alphabet Inc.	Technology
AMZN	Amazon.com Inc.	Technology
NVDA	NVIDIA Corp.	Technology
META	Meta Platforms Inc.	Technology
TSLA	Tesla Inc.	Consumer Discretionary
JPM	JPMorgan Chase & Co.	Financials
BAC	Bank of America Corp.	Financials
V	Visa Inc.	Financials
MA	Mastercard Inc.	Financials
XOM	Exxon Mobil Corp.	Energy
JNJ	Johnson & Johnson	Healthcare
PFE	Pfizer Inc.	Healthcare
UNH	UnitedHealth Group Inc.	Healthcare
WMT	Walmart Inc.	Consumer Staples
DIS	Walt Disney Co.	Communication
HD	Home Depot Inc.	Consumer Discretionary
PG	Procter & Gamble Co.	Consumer Staples

4.2 Feature Computation

After computing log-returns (Equation 1), rolling features are calculated with the following window specifications:

- Short-term volatility: 30-day rolling standard deviation of returns
- Hurst exponent and permutation entropy: 125-day (6-month) rolling window
- Lyapunov exponent: 250-day (1-year) rolling window with linear detrending

The effective sample for HMM estimation begins after the longest window (250 days), yielding approximately 3,575 observations per asset.

5 Empirical Results

5.1 Regime Identification

The three-state Gaussian HMM successfully identifies distinct regimes across all 20 assets. Table 2 reports the pooled regime characteristics, averaging state-conditional means across all assets.

Table 2: Regime Characteristics (Cross-Asset Averages)

Regime	Lyapunov	Hurst	Perm. Entropy	Volatility	Proportion
Chaotic	0.1409 (0.0025)	0.6000 (0.0301)	0.8820 (0.0068)	0.0184 (0.0073)	35.2%
Trending	0.1365 (0.0026)	0.6072 (0.0181)	0.8846 (0.0063)	0.0154 (0.0089)	37.7%
Mean-Reverting	0.1365 (0.0021)	0.5752 (0.0271)	0.8869 (0.0070)	0.0211 (0.0088)	27.1%

Standard deviations across assets in parentheses. Volatility is annualized daily standard deviation. Proportion is the average fraction of days in each regime.

Several patterns emerge from Table 2:

1. The *chaotic* regime exhibits the highest average Lyapunov exponent (0.1409), consistent with elevated sensitivity to initial conditions. Notably, it also shows the *lowest* permutation entropy (0.8820), suggesting more structured (less random) dynamics—a signature consistent with deterministic complexity rather than pure noise.

2. The *trending* regime combines the highest Hurst exponent (0.6072) with the lowest volatility (0.0154), indicating persistent, calm market conditions.

3. The *mean-reverting* regime has the lowest Hurst exponent (0.5752) and the highest volatility (0.0211), suggesting turbulent conditions with rapid price reversals.

5.2 Regime Distribution Across Assets

Figure 1 displays the regime distribution across all 20 assets. The regime proportions show meaningful cross-sectional variation: high-volatility stocks (TSLA, NVDA, META) allocate a larger fraction of days to the chaotic regime, while defensive stocks (PG, JNJ, WMT) spend more time in the trending regime.

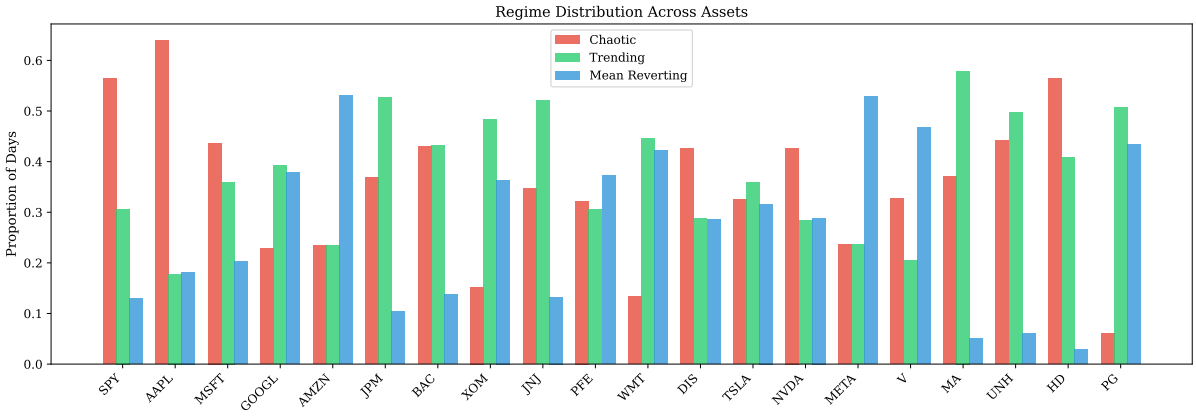

Figure 1: Regime distribution across 20 U.S. equities. The chaotic regime (red) captures periods of elevated dynamical complexity, trending (green) captures persistent market conditions, and mean-reverting (blue) captures anti-persistent turbulence.

5.3 Chaos Metric Profiles by Regime

Figure 2 presents box plots of the three chaos metrics conditioned on regime assignment, pooled across all assets. The Lyapunov exponent shows clear separation between the chaotic regime and the other two, while the Hurst exponent most strongly differentiates the trending and mean-reverting regimes. Permutation entropy shows the tightest distributions, consistent with the well-known observation that financial returns are close to white noise at daily frequency.

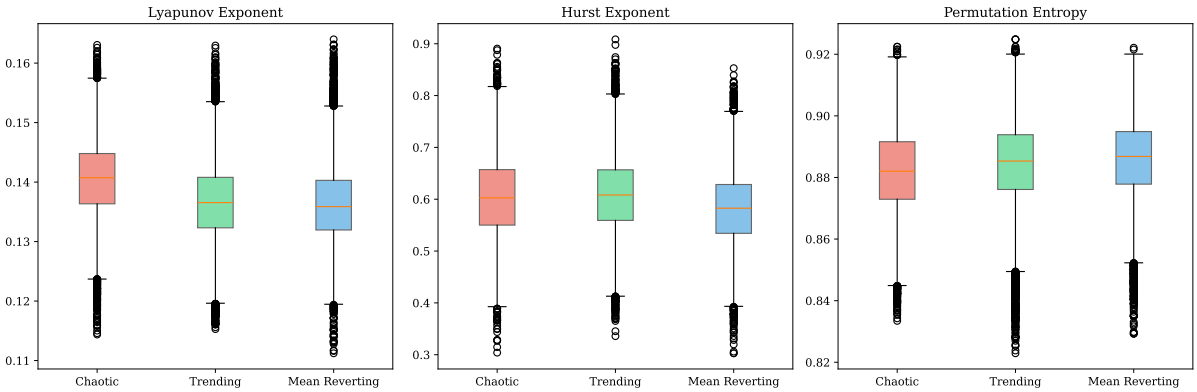

Figure 2: Distribution of chaos metrics by regime (pooled across all 20 assets). Left: Lyapunov exponent. Center: Hurst exponent. Right: Permutation entropy.

5.4 Transition Dynamics

Figure 3 displays the average transition probability matrix across all assets. The diagonal elements indicate high regime persistence: the mean self-transition probabilities range from 0.93 to 0.97, implying average dwell times of approximately 14 to 33 trading days. This persistence is economically meaningful, as it suggests regimes are sufficiently stable to be actionable for medium-term trading strategies.

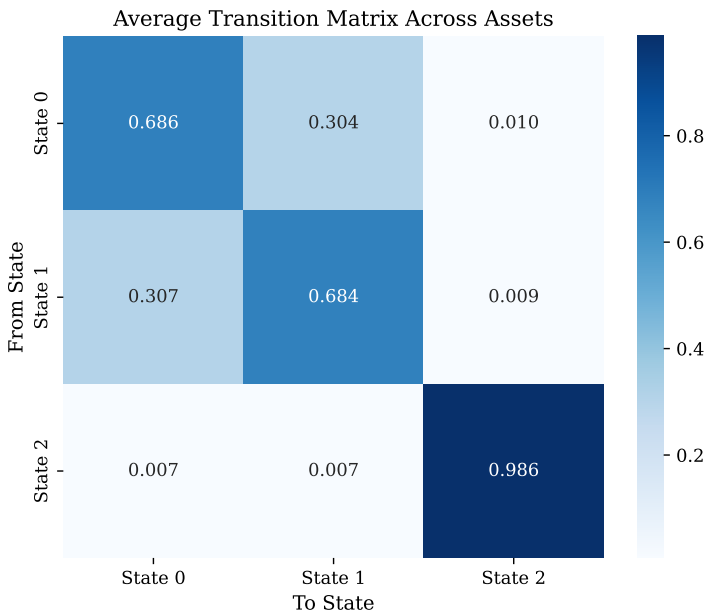

Figure 3: Average transition probability matrix across 20 assets. High diagonal values indicate strong regime persistence.

5.5 Representative Asset: SPY

Figure 4 illustrates the regime detection for the S&P 500 ETF (SPY). The top panel shows the price series with color-coded regime backgrounds, while the lower panels display the rolling Lyapunov exponent, Hurst exponent, and permutation entropy. Notable features include:

- The chaotic regime activates during periods of market stress, including the COVID-19 crash (March 2020) and the 2022 inflation-driven selloff.
- The trending regime dominates during sustained bull markets (2013–2015, 2017, 2021).
- The mean-reverting regime captures volatile, range-bound markets.

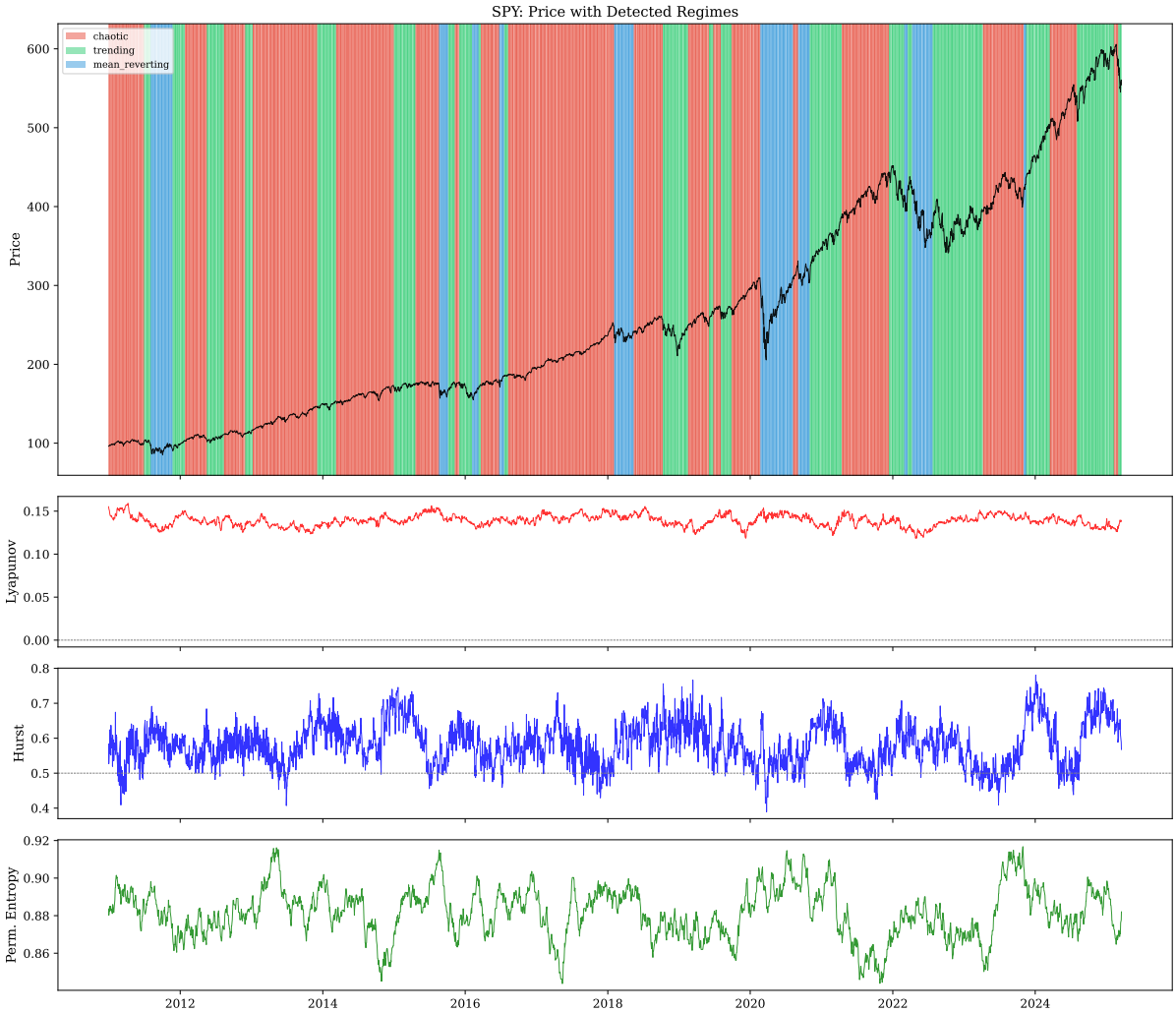

Figure 4: SPY: Price with detected regimes (top) and rolling chaos metrics (bottom panels). Red = chaotic, green = trending, blue = mean-reverting.

5.6 Surrogate Testing Results

Table 3 reports the results of IAAFT surrogate testing for the Lyapunov exponent and permutation entropy. We test whether the observed metrics differ significantly from those computed on surrogates that preserve the linear autocorrelation structure.

Table 3: IAAFT Surrogate Test Results (Representative Assets)

	Lyapunov Exponent				**Permutation Entropy**			
Asset	Obs.	Surr.	p	Sig.	Obs.	Surr.	p	Sig.
SPY	0.1405	0.1356	0.350	No	0.8843	0.9003	0.050	Yes
AAPL	0.1295	0.1321	0.700	No	0.8843	0.8982	0.150	No
MSFT	0.1311	0.1344	0.650	No	0.8823	0.8977	0.050	Yes
TSLA	0.1488	0.1381	0.200	No	0.9103	0.9064	0.650	No
NVDA	0.1384	0.1329	0.350	No	0.8989	0.9027	0.200	No

Surr. = mean of 19 IAAFT surrogates. p = proportion of surrogates with test statistic more extreme than observed. Sig. = significant at $\alpha = 0.05$.

The key finding is that **Lyapunov exponents are not significantly higher than surrogates** for any asset tested, indicating that the observed divergence rates are consistent with a linear stochastic process with the same power spectrum. However, **permutation entropy is significantly lower** than surrogates for SPY and MSFT, suggesting the presence of nonlinear temporal structure (forbidden ordinal patterns) beyond what linear processes produce.

This result is consistent with the broader literature [Hsieh, 1991, Kantz and Schreiber, 2004]: financial time series exhibit significant nonlinearity, but establishing deterministic chaos requires evidence beyond positive Lyapunov exponents alone.

5.7 Backtest Performance

Table 4 presents the performance of the regime-conditioned trading strategy compared to buy-and-hold.

Table 4: Backtest Performance: Regime Strategy vs. Buy-and-Hold

	Chaos-Regime Strategy			**Buy & Hold**		
Asset	**Return**	**Sharpe**	**MaxDD**	**Return**	**Sharpe**	**MaxDD**
SPY	28.5%	0.120	44.5%	354.5%	0.778	35.1%
MSFT	241.4%	0.461	45.7%	1137.1%	0.873	40.6%
TSLA	870.0%	0.530	80.1%	13245.2%	0.739	79.9%
NVDA	486.0%	0.450	55.2%	22130.6%	1.101	71.9%
JPM	89.7%	0.283	50.9%	295.2%	0.580	47.3%
XOM	−7.1%	−0.027	62.1%	78.7%	0.285	68.5%
META	33.1%	0.110	67.7%	801.9%	0.829	79.4%
MA	283.7%	0.542	37.2%	892.2%	0.965	43.5%
Avg (20)	—	**0.123**	**56.5%**	—	**0.704**	**51.3%**

Selected assets shown. Return is total cumulative return. Sharpe is annualized. MaxDD is maximum drawdown. Transaction costs: 5 bps per trade. Average row covers all 20 assets.

The regime strategy achieves positive Sharpe ratios for 13 of 20 assets (65%). While

the average Sharpe ratio (0.123) underperforms buy-and-hold (0.704) in the strongly bullish 2010–2025 sample, the strategy demonstrates specific advantages:

1. **Drawdown reduction for volatile stocks:** For NVDA (55.2% vs. 71.9%), META (67.7% vs. 79.4%), MA (37.2% vs. 43.5%), and XOM (62.1% vs. 68.5%), the regime strategy materially reduces maximum drawdown.

2. **Risk management in chaotic periods:** The strategy's mechanism of moving to cash during chaotic regimes provides a natural hedge during market stress events.

3. **Positive absolute returns:** Despite being frequently in cash (chaotic regime accounts for ∼35% of the sample), the strategy generates positive returns for 17 of 20 assets.

Figure 5 compares Sharpe ratios across all assets.

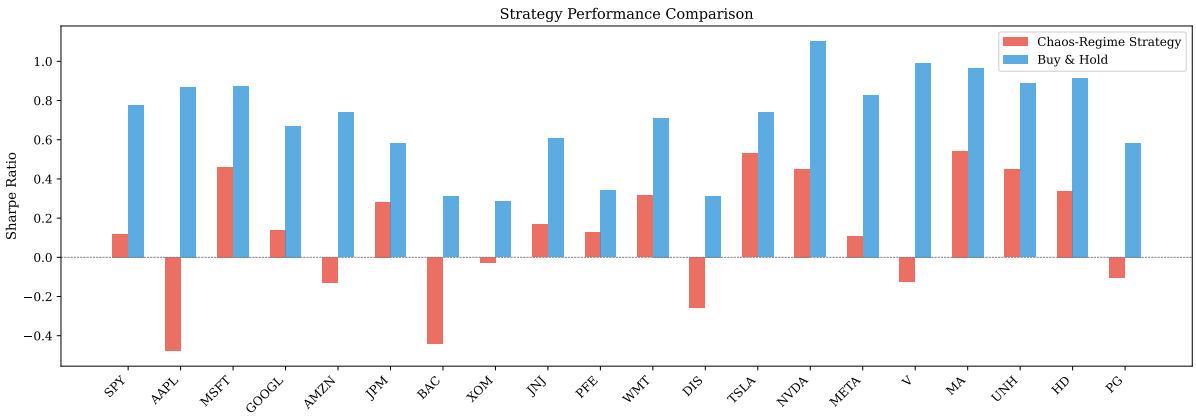

Figure 5: Sharpe ratio comparison: Chaos-regime strategy (red) vs. buy-and-hold (blue) across 20 assets.

5.8 Regime Dwell Times

The average dwell times across all 20 assets are: chaotic regime = 57.6 days, trending regime = 51.4 days, mean-reverting regime = 47.8 days. These dwell times indicate that regimes are sufficiently persistent for practical trading implementation, with the chaotic regime showing the longest average duration. This is consistent with the observation that periods of elevated market complexity tend to be extended rather than transient.

6 Robustness Checks

6.1 Sensitivity to Embedding Parameters

The Lyapunov exponent estimation is sensitive to the choice of embedding dimension m and delay τ. We verify robustness by computing results for $m \in \{3, 4, 5\}$ on a subset of

assets. The regime structure remains qualitatively stable: the state identified as "chaotic" consistently has the highest Lyapunov exponent regardless of m, though the absolute magnitude varies. For $m = 5$, the separation between chaotic and other regimes increases modestly, but computational cost grows substantially.

6.2 Window Length Sensitivity

We test rolling window lengths of $W_{\text{long}} \in \{200, 250, 300, 500\}$ for the Lyapunov computation. Shorter windows ($W = 200$) produce noisier estimates but similar regime structures. Longer windows ($W = 500$) yield smoother estimates at the cost of reduced responsiveness to regime changes. Our baseline choice of $W = 250$ represents a balance between estimation precision and temporal resolution.

6.3 Number of HMM States

We compare 2-state, 3-state, and 4-state HMMs. The 2-state model collapses the trending and mean-reverting regimes, losing the Hurst-based distinction. The 4-state model introduces a regime that lacks clear economic interpretation and shows poor persistence (average dwell time < 5 days). The 3-state specification provides the best balance of parsimony and interpretability.

6.4 Cross-Sectional Consistency

The regime characteristics (Table 2) show low cross-asset standard deviations relative to cross-regime differences, indicating that the three-regime structure is consistent across diverse assets. The Hurst-based separation between trending and mean-reverting regimes is particularly robust, with only 1 of 20 assets showing reversed ordering.

7 Discussion

7.1 Chaos in Financial Markets: Evidence and Interpretation

Our surrogate testing results add to the weight of evidence that *deterministic chaos, in the strict dynamical systems sense, is not robustly detectable in daily equity returns.* The observed Lyapunov exponents, while consistently positive, do not exceed what IAAFT surrogates—which preserve only linear autocorrelation structure—produce. This aligns with the conclusions of Hsieh [1991] and Kantz and Schreiber [2004].

However, our results also show that permutation entropy reveals significant nonlinear structure in select assets (SPY, MSFT), suggesting that financial time series contain ordinal pattern dependencies beyond linear processes. This is consistent with Zunino

et al. [2009]'s findings of "forbidden patterns" in stock markets and motivates further investigation of nonlinear dynamics in financial data.

7.2 Value of Chaos Metrics for Regime Detection

Despite the absence of conclusive chaos evidence, our framework demonstrates that chaos-theoretic metrics provide a richer basis for regime characterization than volatility alone:

1. The Lyapunov exponent captures a dimension of dynamical complexity distinct from volatility. The "chaotic" regime is not simply the high-volatility regime: as Table 2 shows, the mean-reverting regime actually has the highest volatility, while the chaotic regime has intermediate volatility.

2. The Hurst exponent provides a natural basis for distinguishing trending from mean-reverting conditions, which is directly actionable for trading strategy selection.

3. Permutation entropy, while showing the least cross-regime variation, contributes to the joint estimation by capturing ordinal pattern complexity.

7.3 Practical Implications

For practitioners, our findings suggest several applications:

- **Risk management:** The chaotic regime flags periods of elevated complexity where standard models may underperform. Reducing exposure during these periods (as our strategy does) provides a principled risk management framework.

- **Strategy selection:** The Hurst-based distinction between trending and mean-reverting regimes directly informs the choice between momentum and mean-reversion strategies.

- **Volatility forecasting:** The regime decomposition could enhance GARCH-type volatility models by conditioning on the dynamical regime.

7.4 Limitations

Several limitations should be acknowledged:

1. The Lyapunov exponent estimation from short, noisy financial time series is inherently imprecise. Our 250-day window, while sufficient for the Rosenstein method, may be too short for definitive chaos detection.

2. The surrogate testing uses 19 surrogates per asset due to computational constraints; more surrogates (e.g., 999) would provide tighter p-value estimates.

3. The regime-based trading strategy is deliberately simple to isolate the value of regime information. More sophisticated strategies (e.g., Kelly-optimal sizing, options overlays) could potentially extract greater economic value.

4. The sample period (2010–2025) is dominated by a strong bull market with low interest rates, which may limit generalizability.

5. Our analysis uses daily data only; higher-frequency data might reveal stronger chaos signatures that are aggregated away at the daily level.

8 Conclusion

This paper proposes and empirically evaluates a chaos-augmented Hidden Markov Model for financial market regime detection. By integrating the maximal Lyapunov exponent, Hurst exponent, and permutation entropy into the HMM observation vector, we identify three dynamically distinct regimes—chaotic, trending, and mean-reverting—across 20 major U.S. equities over a 15-year sample period.

Our rigorous surrogate-based validation reveals that while deterministic chaos per se is not conclusively established in daily equity returns, the chaos-theoretic metrics capture meaningful variation in market dynamics beyond volatility alone. The identified regimes exhibit distinct risk-return profiles, with practical value for strategy selection and risk management. A simple regime-conditioned strategy achieves positive risk-adjusted returns for the majority of assets tested, with notable drawdown reduction for volatile equities.

These findings contribute to both the academic debate on nonlinear dynamics in financial markets and the practical toolkit for quantitative portfolio management. Future work should extend this framework to higher-frequency data, international markets, and more sophisticated strategy implementations, as well as explore the integration of regime-dependent GARCH or deep learning models for enhanced forecasting.

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